

RANIT PAL

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Education

Indraprastha Institute of Information Technology Delhi(IIT Delhi) 08/2022- 05/2024
M.Tech in Computer Science CGPA, 8.35

Asansol Engineering College 08/2018- 05/2022
B.Tech in Computer Science CGPA, 8.79

Relevant Coursework

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|----------------------|-------------------------|-------------------------|---------------------------|
| • Data Structure | Healthcare | • Machine Learning for | • Artificial Intelligence |
| • Machine Learning | • Information Retrieval | Biomedical Applications | • Reinforcement Learning |
| • Big Data Mining in | • Deep Learning | | • System Programming |

Experience

HCLTech 2024 – present

Data Scientist | Python, LLMops,Pinecone, SageMaker, AWS Bedrock, Fargate, ChromaDB, MCP *Noida*

- Developed a **DQN**-based rApp for real-time energy optimisation in Open RAN, leveraging live Viavi data and insights from the paper "Energy Efficiency in Open RAN: RF Channel Reconfiguration Use Case".
- **Finetuned** Mistral-7B on robotics data to generate accurate, context-aware robot transcripts
- Built an Agentic AI SmartDesk Assistant using **MCP**, deployed on **AWS** with **RAG, Bedrock, Lambda**, and vector search for autonomous document processing and action execution
- Built a SuperApp using **Agentic AI (CrewAI)** to autonomously orchestrate rApps, dynamically managing their lifecycle (start/stop) based on real-time network requirements, and deployed on **AWS (EC2, ECR)** for cloud execution.

API Developer Intern 01/2023 – 05/2023

Research Intern *IIT Delhi*

- Developed multiple APIs for a **Recipe DB** website operated by CoSyLab. Designed and deployed robust RESTful APIs using Spring Boot, Postman and MongoDB to enable users to discover recipes based on region, country.

Machine Learning Intern 08/2023 – 05/2024

Research Intern *IIT Delhi*

- Collected and integrated data from diverse sources to predict phosphorylated sites, applying feature engineering techniques such as **KNN** and **PCA** to enhance model accuracy from 80.6% to 89.8% and the F1 score from 0.72 to 0.88.

Projects

Summarise and Recommend: A BERT-Based Hybrid NLP Recommender System | [GitHub](#) 02/2023–05/2023

- Implemented content-based filtering using research paper titles and abstracts; enhanced performance via section-wise Abstractive and extractive summarisation using Python, **TensorFlow**, **Scikit-learn**, **NLTK**.

Twisafe: NLP-Powered Abusive Tweet Detector | [GitHub](#) 11/2023–12/2023

- Classified Twitter comments using **SVM**, **Voting Classifier** and other ML models; achieved 90% accuracy and 0.92 F1-score using Python, Scikit-learn, TensorFlow, NLTK, NLP, and Git/GitHub.

Telegram ChatBot | *Mistral-7B, QLoRA, SerpAPI, Prompt Handling, Asynchronous Programming* 10/2023–11/2023

- Built an intelligent Telegram bot using the Mistral 7B model optimised via **QLoRA**. Integrated **SerpAPI** for real-time queries and implemented contextual memory, dynamic commands for seamless and human-like interaction.

FastAPI-Based GenAI Document Analyzer with FAISS and LangChain | *LLaMA3-8B, QLoRA* 9/2023–11/2023

- Built a GenAI app with **LLaMA3-8B**, **LangChain**, **FAISS** and **FastAPI** to auto-generate contextual Q&A from uploaded documents for advanced content understanding.

Technical Skills

Languages: Python, C/C++, Java, HTML/CSS, SQL

Tools/Frameworks: Python Libraries, VS Code, MySQL, Git/GitHub, Spring Boot, MongoDB, Linux, Docker, Kubernetes, AWS, Azure, Hugging Face

AI/ML: TensorFlow, PyTorch, MCP, Keras, ScikitLearn, Pandas, NumPy, SciPy, NLTK, LlamaIndex, LLMs(Claude, LLaMa, GPT), Agentic AI, Generative AI (**RAG**, LangChain, LangGraph, LoRA, QLoRA, Finetune LLMs, Transformer)

Achievements and Leadership Role

- Secured **AIR 19 in PGCAT** and qualified **JEE and GATE CS 2021** (B.Tech 3rd year) and GATE CS 2022.
- Lead presenter for HCL at **AWS international Hackathon**; built a SuperApp with Agentic AI (CrewAI) on AWS.

Key competencies

- Strong problem-solving skills, effective analytical thinking, solid understanding of software development principles, teamwork, communication, adaptability and time management.